

Julian Rosenberger — Curriculum Vitae

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Education

- Sept 2022 – Mar 2027 **PhD in Information Systems**, TU Dresden (expected defense Mar 2027)
Dissertation: *The Role of Interpretability in Human–AI Collaboration*
First supervisor: Patrick Zschech Second supervisor: Florian Siems
- Sept 2021 – Jan 2022 **Exchange Semester (Computer Science)**, University of Southern Denmark, Odense
- Oct 2020 – Mar 2023 **MSc International Information Systems**, FAU Erlangen–Nuremberg; grade 1.21, with distinction
Thesis: *Interpretable Machine Learning for Length-of-Stay and Mortality Prediction in ICUs – From Understanding to Achieving Model Interpretability* (grade 1.0)
- Oct 2015 – Sep 2018 **BA Economics (Insurance)**, DHBW Mannheim, dual programme with Allianz Beratungs- und Vertriebs-AG; grade 1.6
Thesis: *Acceptance of AI in the Form of Digital Assistants in the Insurance Industry* (grade 1.4)

Employment

- since Jul 2026 **Doctoral Researcher**, Chair of Business Information Systems, esp. Intelligent Systems and Services, TU Dresden. Prof. Patrick Zschech
Research on human–AI collaboration with agentic AI (procedural transparency and verification behavior). Planned interview study with developers at Soloplan GmbH on coding agents. Teaching in programming and agentic AI; supervision of 8+ theses.
- May 2024 – Jul 2026 **Doctoral Researcher**, Chair for Explainable AI in Business Value Creation, University of Regensburg. Prof. Mathias Kraus
Pre-registered user studies ($N > 250$) on transparency, adjustability, and personalisation in interpretable ML. Development of GAMs, contextual bandits, and LLM-based conversational explainability systems.
- Sept 2022 – May 2024 **Doctoral Researcher**, Junior Research Group on White-box AI, FAU Erlangen–Nuremberg. Prof. Mathias Kraus, Prof. Patrick Zschech
Initiated doctoral research on interpretable ML, human–AI collaboration, and trust calibration. Conducted behavioral experiments.
- Mar 2021 – Aug 2022 **Research Assistant**, FAU Erlangen–Nuremberg
Computer vision for defect detection in images from weaving production; developed a course on NLP methods and applications. Supported “Introduction to Computer Science” (80+ students); designed tutorials on programming fundamentals based on Harvard’s CS50.
- Apr 2020 – Apr 2021 **Product Design & UX Research**, Allianz Technology (Global Digital Factory), Munich
UX research, design tool evaluation, wireframe-to-screen workflows. Product Owner for “Allianz Future App”.
- Oct 2018 – Feb 2020 **Technical Sales Manager**, Hufschmied Zerspanungssysteme GmbH, Bobingen
Won and advised B2B clients in Austria and southern Bavaria; planned and delivered customer projects.

Publications

- 2026 Bohlen, L., J. Rosenberger, N. Hambauer, D. Zähringer, V. Franz, P. Zschech, and M. Kraus. *Toward Generalizable and Interpretable Machine Learning Models in Healthcare: Insights from ICU Outcome Predictions*. *Health Care Management Science* 29.2, p. 23.
- 2026 Kruschel, S., J. Rosenberger, L. Bohlen, M. Kraus, and P. Zschech. *Can Conversational XAI Improve User Performance? An Experimental Study*. *Proceedings of the 34th European Conference on Information Systems (ECIS)*. Milan, Italy.
- 2026 Rosenberger, J., S. Kuhlemann, V. Tiefenbeck, M. Kraus, and P. Zschech. *Understanding Data-Sharing with AI Systems: The Roles of Transparency, Trust, and the Processing Entity*. *Pacific Asia Journal of the Association for Information Systems* 18.2, pp. 1–26.
- 2025 Bohlen, L., S. Kruschel, J. Rosenberger, P. Zschech, and M. Kraus. *Overcoming Algorithm Aversion with Transparency: Can Transparent Predictions Change User Behavior?* *Wirtschaftsinformatik 2025 Proceedings*.
- 2025 Bohlen, L., J. Rosenberger, P. Zschech, and M. Kraus. *Leveraging Interpretable Machine Learning in Intensive Care*. *Annals of Operations Research* 347.2, pp. 1093–1132.
- 2025 Gümmer, P., J. Rosenberger, M. Kraus, P. Zschech, and N. Hambauer. *Unveiling Location-Specific Price Drivers: A Two-Stage Cluster Analysis for Interpretable House Price Predictions*. *Wirtschaftsinformatik 2025 Proceedings*.
- 2025 Kruschel, S., L. Bohlen, J. Rosenberger, P. Zschech, and M. Kraus. *Quantifying Visual Properties of GAM Shape Plots: Impact on Perceived Cognitive Load and Interpretability*. *Proceedings of the 58th Hawaii International Conference on System Sciences*, pp. 1410–1419.
- 2025 Pätz, L., M. Beyer, J. Späth, L. Bohlen, P. Zschech, M. Kraus, and J. Rosenberger. *Analyzing German Parliamentary Speeches: A Machine Learning Approach for Topic and Sentiment Classification*. *Wirtschaftsinformatik 2025 Proceedings*.
- 2025 Rosenberger, J., P. Schröppel, S. Kruschel, M. Kraus, P. Zschech, and M. Förster. *Navigating the Rashomon Effect: How Personalization Can Help Adjust Interpretable Machine Learning Models to Individual Users*. *European Conference on Information Systems (ECIS) 2025 Proceedings*.
- 2025 Rosenberger, J., L. Wolfrum, S. Weinzierl, M. Kraus, and P. Zschech. *Career-BERT: Matching Resumes to ESCO Jobs in a Shared Embedding Space for Generic Job Recommendations*. *Expert Systems with Applications* 275, p. 127043.
- 2024 Hofmann, S., C. Sommermann, M. Kraus, P. Zschech, and J. Rosenberger. *Hate Speech and Sentiment of YouTube Video Comments From Public and Private Sources Covering the Israel-Palestine Conflict*. *Wirtschaftsinformatik 2024 Proceedings*.
- 2024 Rosenberger, J., S. Kuhlemann, V. Tiefenbeck, M. Kraus, and P. Zschech. *The Impact of Transparency in AI Systems on Users' Data-Sharing Intentions: A Scenario-Based Experiment*. *Wirtschaftsinformatik 2024 Proceedings*.

Work in Progress

2026	Shown, therefore overseen? Procedural transparency and the illusion of oversight in agentic AI (target: ECIS 2027) Overreliance and illusions of understanding in AI-assisted decision-making The role of explanation scope and representational format in human–AI collaboration
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Grant Proposals (submitted)

2026	MARS-KI (BMFTR): multi-agent market research system for SMEs. Lead of proposal development with TU Dresden, a marketing chair, and DH Sachsen
2026	MARA (BMFTR, MedTech4Women): clinical decision support for peripartum depression. Lead of the work package on interpretable ML
2026	WEAVE (DFG/FWF): agentic AI and NeuroIS; applicants Patrick Zschech (TU Dresden) and Bernhard Lutz (University of Vienna). Contributed the state of the art sections

Selected Talks & Presentations

Jun 2026	ECIS 2026 , Milan — <i>Can Conversational XAI Improve User Performance? An Experimental Study</i>
Jun 2025	ECIS 2025 , Amman — <i>Navigating the Rashomon Effect: How Personalization Can Help Adjust Interpretable Machine Learning Models</i>
May 2025	OJA Forum (Bertelsmann Stiftung) , Online — <i>Beyond Ad Matching: Using CareerBERT to Map Resumes to ESCO Job Categories</i> (invited talk)
Jan 2025	HICSS 2025 , Hawai'i — <i>Quantifying Visual Properties of GAM Shape Plots</i>
Sept 2024	WI 2024 , Würzburg — <i>The Impact of Transparency in AI Systems on Users' Data-Sharing Intentions</i>

Teaching

Winter 2026/27	Diplom Wirtschaftsinformatik “Programming and Databases” (weekly exercise), TU Dresden
Winter 2026/27	Seminar “Agentic AI: Understanding and Using AI Agents” (new), TU Dresden
Summer 2026	“Gestaltungsansätze der Advanced Data Analytics” (GADA), TU Dresden
Winter 2025/26	MSc Information Systems “Theoretical Seminar”, University of Regensburg
Summer 2025	BSc Information Systems “Data Analytics”, University of Regensburg
Summer 2022	BSc Information Systems “IT-based Process Automation”, FAU Erlangen-Nuremberg
2021 – 2022	MSc International Information Systems “Introduction to Computer Science”, FAU Erlangen-Nuremberg

Awards & Scholarships

2025	ECIS 2025 Doctoral Consortium — Selected participant
2024	Best Student Paper Award, Wirtschaftsinformatik (WI 2024)

Selected Association & Committee Work

2024 – present	Reviewer for ACM Transactions on Information Systems (TOIS), Data Technologies and Applications, ECIS, and Wirtschaftsinformatik (WI)
2024 – 2025	Organizer of White-box AI Research Discussions
2021 – 2022	President of student council “FSI Wirtschaftsinformatik & International Information Systems (IIS)” (FAU)
2020 – 2022	Member of MSc IIS quality circle and FAU’s internationalization committee

Languages & Skills

Languages	German (native); English (fluent, TOEFL iBT 106); French (elementary)
Technical	Python (pandas, scikit-learn, PyTorch, HuggingFace), R (dplyr, ggplot2, lme4), SQL, Git; Claude Code, Codex, OpenCode

Last updated: October 9, 2026